

Math 31 Curriculum

Advanced Algebra | Middlesex School Mathematics

COURSE AT A GLANCE

Review Content

REVIEW

Primary resources: [2.2 Use a Problem Solving Strategy](#); [5.2 Properties of Exponents and Scientific Notation](#); [5.1 Add and Subtract Polynomials](#); [5.3 Multiply Polynomials](#)

UNIT LEARNING OBJECTIVES

I can calculate a percentage, percent increase, or percent decrease.

I can simplify an expression using exponent properties.

I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure.

I can expand a product of two binomials.

Linear Functions and Function Notation

REQUIRED

Primary resources: [College Algebra 2e 3.1 Functions and Function Notation](#); [College Algebra 2e 3.3 Rates of Change and Behavior of Graphs](#); [College Algebra 2e 3.2 Domain and Range](#); [College Algebra 2e 4.1 Linear Functions](#); [Intermediate Algebra 2e 4.1 Solve Systems of Linear Equations with Two Variables](#)

UNIT LEARNING OBJECTIVES

I can interpret function notation as a statement about input and output.

I can determine whether a graph or table represents a function.

I can evaluate a function or solve for an input using a graph, table, or formula.

I can identify independent and dependent variables in a relationship.

I can calculate average rate of change from two points.

I can interpret interval notation and identify increasing or decreasing intervals.

I can determine total change from an average rate of change over an interval.

I can determine and interpret the units of an average rate of change.

I can identify a linear function from a table, graph, or formula.

I can interpret the slope and intercept of a linear model.

I can determine the intersection of two linear relationships graphically or algebraically.

I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point.

Function Behavior and Structure

REQUIRED

Primary resources: [3.1 Functions and Function Notation](#); [3.2 Domain and Range](#); [3.5 Transformation of Functions](#); [3.4 Composition of Functions](#); [3.7 Inverse Functions](#); [3.3 Rates of Change and Behavior of Graphs](#)

UNIT LEARNING OBJECTIVES

I can determine whether a relation is a function.

I can evaluate a function from a formula, graph, or table.

I can determine domain and range from a formula, graph, or table.

I can determine algebraic domain restrictions caused by denominators and even roots.

I can graph a piecewise function from its formula.

I can write a piecewise formula that represents a graph.

I can graph a transformed function using shifts, stretches, compressions, and reflections.

I can describe how a transformation affects domain, range, or asymptotes.

I can evaluate a composition from formulas, graphs, or tables.

I can write a formula for a composition of two functions.

I can find an inverse function algebraically.

I can evaluate and interpret an inverse from a graph, formula, or table.

I can determine and interpret the units of an inverse function.

I can explain how the domain and range of a function relate to those of its inverse.

I can determine the range of a function by analyzing the domain of its inverse.

I can verify algebraically that two functions are inverses.

I can interpret the meaning and units of a composition.

I can classify a graph or table as concave up or concave down using changes in average rate of change.

Function Behavior and Structure (Continued)

REQUIRED

UNIT LEARNING OBJECTIVES - CONTINUED

I can determine whether a function is invertible using the horizontal line test.

I can graph an inverse function as a reflection across the line $y = x$.

Quadratic Functions and Equations

REQUIRED

Primary resources: [5.1 Quadratic Functions](#); [2.5 Quadratic Equations](#)
Supplemental resources: [5.3 Graphs of Polynomial Functions](#); [5.5 Zeros of Polynomial Functions](#)

UNIT LEARNING OBJECTIVES

I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function.

I can connect standard, factored, and vertex forms to features of a quadratic graph.

I can rewrite a quadratic function in a form that reveals a requested feature.

I can solve a quadratic equation by factoring or using the quadratic formula.

I can graph a quadratic function using its algebraic structure and key features.

I can construct a quadratic function from specified zeros, a vertex, or graphical information.

I can create and interpret a quadratic trajectory model.

Exponential Functions and Models

REQUIRED

Primary resources: [6.1 Exponential Functions](#); [6.2 Graphs of Exponential Functions](#)

UNIT LEARNING OBJECTIVES

I can distinguish linear change from exponential change using differences, ratios, or percent change.

I can identify initial value and growth or decay factor in an exponential model.

I can write an exponential function from a table, graph, or context.

I can evaluate and interpret an exponential model.

I can calculate growth under periodic compounding.

I can calculate growth under continuous compounding.

I can determine an exponential formula from two points or other sufficient data.

I can determine domain, range, intercepts, and asymptotes of an exponential function.

I can compare the growth rates of two exponential functions.

I can determine the percent growth or decay represented by a continuous exponential model.

Logarithmic Functions and Models

REQUIRED

Primary resources: [6.3 Logarithmic Functions](#); [6.5 Logarithmic Properties](#); [6.6 Exponential and Logarithmic Equations](#); [6.7 Exponential and Logarithmic Models](#); [6.4 Graphs of Logarithmic Functions](#)
Supplemental resources: [6.2 Graphs of Exponential Functions](#)

UNIT LEARNING OBJECTIVES

I can convert between logarithmic and exponential forms.

I can evaluate a logarithm using its exponential meaning.

I can expand or condense an expression using product, quotient, and power properties.

I can solve an exponential equation using logarithms.

I can solve a logarithmic equation using exponential form and check domain restrictions.

I can calculate and interpret doubling time or half-life.

I can determine domain, range, intercepts, and asymptotes of a logarithmic function.

I can explain how exponential and logarithmic functions undo one another.

I can solve a multi-step exponential or logarithmic equation.

I can determine an intersection of exponential functions graphically or algebraically.

I can rewrite a discrete exponential model as an equivalent continuous exponential model.

I can solve and interpret an exponential or logarithmic modeling problem.

Extension Content

EXTENSION

Primary resources: [3.6 Absolute Value Functions](#); [3.5 Transformation of Functions](#); [6.7 Exponential and Logarithmic Models](#)

UNIT LEARNING OBJECTIVES

I can analyze an absolute-value function using transformations.

I can determine even or odd symmetry from a graph or formula.

I can interpret a contextual logarithmic scale.

IN-DEPTH CURRICULUM

SKILL PROGRESSION



Introduce



Reinforce



Deepen



Apply

Review Content

REVIEW

OpenStax coverage: **Primary resources:** [2.2 Use a Problem Solving Strategy](#); [5.2 Properties of Exponents and Scientific Notation](#); [5.1 Add and Subtract Polynomials](#); [5.3 Multiply Polynomials](#)

M31-REV-001

I can calculate a percentage, percent increase, or percent decrease.

OpenStax: **Primary resources:** [2.2 Use a Problem Solving Strategy](#)

SUPPORTING SKILLS

- Calculate a percentage of a quantity or determine what percent one quantity is of another.
SK-NUM-PERCENT-CALCULATE · inherited from middle school
- Calculate and interpret percent increase or decrease.
SK-NUM-PERCENT-CHANGE · inherited from middle school

M31-REV-002

I can simplify an expression using exponent properties.

OpenStax: **Primary resources:** [5.2 Properties of Exponents and Scientific Notation](#)

SUPPORTING SKILLS

- Rewrite a negative exponent using a reciprocal.
SK-EXP-NEGATIVE · first introduced in Math 21
- Apply the power rule for exponents.
SK-EXP-POWER · first introduced in Math 12
- Apply the product rule for exponents.
SK-EXP-PRODUCT · first introduced in Math 12
- Apply the quotient rule for exponents.
SK-EXP-QUOTIENT · first introduced in Math 12
- Apply the zero-exponent rule.
SK-EXP-ZERO · first introduced in Math 21

M31-REV-003

I can use polynomial vocabulary to distinguish terms, factors, coefficients, degree, and polynomial structure.

OpenStax: Primary resources: [5.1 Add and Subtract Polynomials](#)

SUPPORTING SKILLS

- R** Identify terms, factors, coefficients, variables, and degree in a polynomial expression.
SK-POLY-VOCABULARY · first introduced in Math 12

M31-REV-004

I can expand a product of two binomials.

OpenStax: Primary resources: [5.3 Multiply Polynomials](#)

SUPPORTING SKILLS

- A** Apply the distributive property in either direction.
SK-ALG-DISTRIBUTE · inherited from middle school
- R** Multiply two binomials using distribution.
SK-POLY-BINOMIAL-MULTIPLY · first introduced in Math 12

Linear Functions and Function Notation

REQUIRED

OpenStax coverage: **Primary resources:** [College Algebra 2e 3.1 Functions and Function Notation](#); [College Algebra 2e 3.3 Rates of Change and Behavior of Graphs](#); [College Algebra 2e 3.2 Domain and Range](#); [College Algebra 2e 4.1 Linear Functions](#); [Intermediate Algebra 2e 4.1 Solve Systems of Linear Equations with Two Variables](#)

M31-LIN-001

I can interpret function notation as a statement about input and output.

OpenStax: **Primary resources:** [3.1 Functions and Function Notation](#)

SUPPORTING SKILLS

- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A** Identify the input and output represented by function notation.
SK-FUNC-NOTATION-READ · first introduced in Math 12
- I** Use function notation to express and solve an input-output question.
SK-FUNC-NOTATION-SOLVE

M31-LIN-002

I can determine whether a graph or table represents a function.

OpenStax: **Primary resources:** [3.1 Functions and Function Notation](#)

SUPPORTING SKILLS

- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D** Apply the vertical line test to a graph.
SK-FUNC-VERTICAL-LINE · first introduced in Math 12

M31-LIN-003

I can evaluate a function or solve for an input using a graph, table, or formula.

OpenStax: **Primary resources:** [3.1 Functions and Function Notation](#)

SUPPORTING SKILLS

- A** Use inverse operations to maintain an equivalent equation.
SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A** Use function notation to express and solve an input-output question.
SK-FUNC-NOTATION-SOLVE · first introduced in Math 31
- D** Solve for an input associated with a specified output.
SK-FUNC-SOLVE-INPUT · first introduced in Math 21

M31-LIN-004

I can identify independent and dependent variables in a relationship.

OpenStax: **Primary resources:** [3.1 Functions and Function Notation](#)

SUPPORTING SKILLS

- I** Identify independent and dependent variables.
SK-FUNC-INDEPENDENT-DEPENDENT
- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12

M31-LIN-005

I can calculate average rate of change from two points.

OpenStax: **Primary resources:** [3.3 Rates of Change and Behavior of Graphs](#)

SUPPORTING SKILLS

- D** Calculate average rate of change over an interval.
SK-FUNC-AROC · first introduced in Math 21
- A** Calculate slope from two points.
SK-LIN-SLOPE-POINTS · first introduced in Math 12

M31-LIN-006

I can interpret interval notation and identify increasing or decreasing intervals.

OpenStax: **Primary resources:** [3.2 Domain and Range](#)

SUPPORTING SKILLS

- I** Describe increasing or decreasing behavior over intervals.
SK-FUNC-INTERVAL-BEHAVIOR
- D** Translate between inequality and interval notation.
SK-NOT-INTERVAL · first introduced in Math 12

M31-LIN-007

I can determine total change from an average rate of change over an interval.

OpenStax: **Primary resources:** [3.3 Rates of Change and Behavior of Graphs](#)

SUPPORTING SKILLS

- A** Calculate average rate of change over an interval.
SK-FUNC-AROC · first introduced in Math 21
- I** Use average rate of change and interval length to calculate total change.
SK-FUNC-AROC-TOTAL-CHANGE
- A** Add, subtract, multiply, and divide signed integers.
SK-NUM-INTEGGER-OPERATE · inherited from middle school

M31-LIN-008

I can determine and interpret the units of an average rate of change.

OpenStax: **Primary resources:** [3.3 Rates of Change and Behavior of Graphs](#)

SUPPORTING SKILLS

- I** Determine rate units from input- and output-axis units.
SK-FUNC-AROC-UNITS
- D** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M31-LIN-009

I can identify a linear function from a table, graph, or formula.

OpenStax: **Primary resources:** [4.1 Linear Functions](#)

SUPPORTING SKILLS

- I** Use constant differences or ratios to distinguish linear and exponential patterns.
SK-EXP-DIFFERENCE-RATIO
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21
- D** Use tabular patterns to predict graph shape and behavior.
SK-REP-TABLE-GRAPH · first introduced in Math 21

M31-LIN-010

I can interpret the slope and intercept of a linear model.

OpenStax: **Primary resources:** [4.1 Linear Functions](#)

SUPPORTING SKILLS

- A** Plot an intercept accurately on a coordinate plane.
SK-LIN-PLOT-INTERCEPT · first introduced in Math 12
- A** Determine slope from a graph.
SK-LIN-SLOPE-GRAPH · first introduced in Math 12
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M31-LIN-011

I can determine the intersection of two linear relationships graphically or algebraically.

OpenStax: **Primary resources:** [4.1 Solve Systems of Linear Equations with Two Variables](#)

SUPPORTING SKILLS

- A** Check an ordered pair in both equations of a system.
SK-SYS-CHECK · first introduced in Math 12
- A** Add equations to eliminate a variable.
SK-SYS-ELIMINATE · first introduced in Math 12
- A** Locate and interpret the intersection of two lines.
SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12
- A** Substitute an equivalent expression for a variable.
SK-SYS-SUBSTITUTE · first introduced in Math 12

M31-LIN-012

I can write equations of horizontal, vertical, parallel, and perpendicular lines through a specified point.

OpenStax: **Primary resources:** [4.1 Linear Functions](#)

SUPPORTING SKILLS

- A** Translate quantities and relationships into an equation.
SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- A** Convert among common forms of a linear equation.
SK-LIN-FORM-CONVERT · first introduced in Math 12
- D** Recognize that distinct parallel lines have equal slopes.
SK-LIN-PARALLEL-SLOPE · first introduced in Math 12
- D** Determine the negative reciprocal of a nonzero slope.
SK-LIN-PERPENDICULAR-SLOPE · first introduced in Math 12

Function Behavior and Structure

REQUIRED

OpenStax coverage: **Primary resources:** [3.1 Functions and Function Notation](#); [3.2 Domain and Range](#); [3.5 Transformation of Functions](#); [3.4 Composition of Functions](#); [3.7 Inverse Functions](#); [3.3 Rates of Change and Behavior of Graphs](#)

M31-FUN-001

I can determine whether a relation is a function.

OpenStax: **Primary resources:** [3.1 Functions and Function Notation](#)

SUPPORTING SKILLS

- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A** Apply the vertical line test to a graph.
SK-FUNC-VERTICAL-LINE · first introduced in Math 12

M31-FUN-002

I can evaluate a function from a formula, graph, or table.

OpenStax: **Primary resources:** [3.1 Functions and Function Notation](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- D** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A** Solve for an input associated with a specified output.
SK-FUNC-SOLVE-INPUT · first introduced in Math 21

M31-FUN-003

I can determine domain and range from a formula, graph, or table.

OpenStax: **Primary resources:** [3.2 Domain and Range](#)

SUPPORTING SKILLS

- I** Determine domain restrictions from denominators, even roots, and logarithms.
SK-FUNC-DOMAIN-FORMULA
- D** Read domain and range from a graph.
SK-FUNC-DOMAIN-RANGE-GRAPH · first introduced in Math 12
- D** Read domain and range from a table.
SK-FUNC-DOMAIN-RANGE-TABLE · first introduced in Math 12

M31-FUN-004

I can determine algebraic domain restrictions caused by denominators and even roots.

OpenStax: **Primary resources:** [3.2 Domain and Range](#)

SUPPORTING SKILLS

- A** Factor a quadratic trinomial with a nonunit leading coefficient.
SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A** Factor a monic quadratic trinomial.
SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- D** Require an even-root radicand to be nonnegative when determining domain.
SK-FUNC-DOMAIN-EVEN-ROOT · first introduced in Math 21
- D** Determine domain restrictions from denominators, even roots, and logarithms.
SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- A** Determine excluded values from an original denominator.
SK-RAT-RESTRICTIONS · first introduced in Math 21

M31-FUN-005

I can graph a piecewise function from its formula.

OpenStax: **Primary resources:** [3.2 Domain and Range](#)

SUPPORTING SKILLS

- A** Describe increasing or decreasing behavior over intervals.
SK-FUNC-INTERVAL-BEHAVIOR · first introduced in Math 31
- I** Select the correct rule of a piecewise function for a given input.
SK-FUNC-PIECEWISE-EVALUATE
- I** Graph each rule on its stated domain with correct endpoints.
SK-FUNC-PIECEWISE-GRAPH

M31-FUN-006

I can write a piecewise formula that represents a graph.

OpenStax: **Primary resources:** [3.2 Domain and Range](#)

SUPPORTING SKILLS

- A** Describe increasing or decreasing behavior over intervals.
SK-FUNC-INTERVAL-BEHAVIOR · first introduced in Math 31
- I** Translate a segmented graph into a piecewise formula.
SK-FUNC-PIECEWISE-WRITE
- A** Convert among common forms of a linear equation.
SK-LIN-FORM-CONVERT · first introduced in Math 12

M31-FUN-007

I can graph a transformed function using shifts, stretches, compressions, and reflections.

OpenStax: **Primary resources:** [3.5 Transformation of Functions](#)

SUPPORTING SKILLS

- I** Interpret transformations applied to function inputs.
SK-FUNC-TRANSFORM-HORIZONTAL
- I** Distinguish stretches from compressions and horizontal effects from vertical effects.
SK-FUNC-TRANSFORM-SCALE
- I** Interpret transformations applied to function outputs.
SK-FUNC-TRANSFORM-VERTICAL
- A** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-FUN-008

I can describe how a transformation affects domain, range, or asymptotes.

OpenStax: **Primary resources:** [3.5 Transformation of Functions](#)

SUPPORTING SKILLS

- I** Exchange domain and range between a function and its inverse.
SK-FUNC-INVERSE-DOMAIN-RANGE
- D** Interpret transformations applied to function inputs.
SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31
- D** Distinguish stretches from compressions and horizontal effects from vertical effects.
SK-FUNC-TRANSFORM-SCALE · first introduced in Math 31
- D** Interpret transformations applied to function outputs.
SK-FUNC-TRANSFORM-VERTICAL · first introduced in Math 31

M31-FUN-009

I can evaluate a composition from formulas, graphs, or tables.

OpenStax: **Primary resources:** [3.4 Composition of Functions](#)

SUPPORTING SKILLS

- I** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA
- I** Follow intermediate values to compose functions from tables.
SK-FUNC-COMPOSE-TABLE
- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- I** Read and write notation for function composition and inverse functions.
SK-FUNC-STRUCTURE-NOTATION

M31-FUN-010

I can write a formula for a composition of two functions.

OpenStax: **Primary resources:** [3.4 Composition of Functions](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- D** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31

M31-FUN-011

I can find an inverse function algebraically.

OpenStax: **Primary resources:** [3.7 Inverse Functions](#)

SUPPORTING SKILLS

- A** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- I** Exchange variables and solve to obtain an inverse formula.
SK-FUNC-INVERSE-ALGEBRA
- I** Exchange input and output values when interpreting an inverse.
SK-FUNC-INVERSE-SWAP
- A** Read and write notation for function composition and inverse functions.
SK-FUNC-STRUCTURE-NOTATION · first introduced in Math 31

M31-FUN-012

I can evaluate and interpret an inverse from a graph, formula, or table.

OpenStax: **Primary resources:** [3.7 Inverse Functions](#)

SUPPORTING SKILLS

- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- A** Exchange variables and solve to obtain an inverse formula.
SK-FUNC-INVERSE-ALGEBRA · first introduced in Math 31
- D** Exchange input and output values when interpreting an inverse.
SK-FUNC-INVERSE-SWAP · first introduced in Math 31

M31-FUN-013

I can determine and interpret the units of an inverse function.

OpenStax: **Primary resources:** [3.7 Inverse Functions](#)

SUPPORTING SKILLS

- I** Reverse input and output units when interpreting an inverse.
SK-FUNC-INVERSE-UNITS
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M31-FUN-014

I can explain how the domain and range of a function relate to those of its inverse.

OpenStax: **Primary resources:** [3.7 Inverse Functions](#)

SUPPORTING SKILLS

- I** Exchange domain and range between a function and its inverse.
SK-FUNC-INVERSE-DOMAIN-RANGE
- A** Exchange input and output values when interpreting an inverse.
SK-FUNC-INVERSE-SWAP · first introduced in Math 31

M31-FUN-015

I can determine the range of a function by analyzing the domain of its inverse.

OpenStax: **Primary resources:** [3.7 Inverse Functions](#)

SUPPORTING SKILLS

- A** Determine domain restrictions from denominators, even roots, and logarithms.
SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- D** Exchange domain and range between a function and its inverse.
SK-FUNC-INVERSE-DOMAIN-RANGE · first introduced in Math 31

M31-FUN-016

I can verify algebraically that two functions are inverses.

OpenStax: **Primary resources:** [3.7 Inverse Functions](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- A** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- I** Verify inverse functions using composition.
SK-FUNC-INVERSE-VERIFY

M31-FUN-017

I can interpret the meaning and units of a composition.

OpenStax: **Primary resources:** [3.4 Composition of Functions](#)

SUPPORTING SKILLS

- D** Substitute one function formula into another.
SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- D** Follow intermediate values to compose functions from tables.
SK-FUNC-COMPOSE-TABLE · first introduced in Math 31
- A** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- D** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M31-FUN-018

I can classify a graph or table as concave up or concave down using changes in average rate of change.

OpenStax: **Primary resources:** [3.3 Rates of Change and Behavior of Graphs](#)

SUPPORTING SKILLS

- A** Calculate average rate of change over an interval.
SK-FUNC-AROC · first introduced in Math 21
- I** Use successive average rates of change to classify concavity.
SK-FUNC-CONCAVITY-AROC
- D** Use tabular patterns to predict graph shape and behavior.
SK-REP-TABLE-GRAPH · first introduced in Math 21

M31-FUN-019

I can determine whether a function is invertible using the horizontal line test.

OpenStax: **Primary resources:** [3.7 Inverse Functions](#)

SUPPORTING SKILLS

- I** Apply the horizontal line test for invertibility.
SK-FUNC-INVERSE-HLT
- I** Test whether a function is one-to-one on a domain.
SK-FUNC-ONE-TO-ONE

M31-FUN-020

I can graph an inverse function as a reflection across the line $y = x$.

OpenStax: **Primary resources:** [3.7 Inverse Functions](#)

SUPPORTING SKILLS

- A** Exchange domain and range between a function and its inverse.
SK-FUNC-INVERSE-DOMAIN-RANGE · first introduced in Math 31
- D** Exchange input and output values when interpreting an inverse.
SK-FUNC-INVERSE-SWAP · first introduced in Math 31
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

Quadratic Functions and Equations

REQUIRED

OpenStax coverage: **Primary resources:** [5.1 Quadratic Functions](#); [2.5 Quadratic Equations](#)

Supplemental resources: [5.3 Graphs of Polynomial Functions](#); [5.5 Zeros of Polynomial Functions](#)

M31-QUAD-001

I can identify the vertex, intercepts, axis of symmetry, and concavity of a quadratic function.

OpenStax: **Primary resources:** [5.1 Quadratic Functions](#)

SUPPORTING SKILLS

- A** Use successive average rates of change to classify concavity.
SK-FUNC-CONCAVITY-AROC · first introduced in Math 31
 - A** Identify standard, factored, and vertex forms.
SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
 - D** Determine x- and y-intercepts of a quadratic function.
SK-QUAD-INTERCEPTS · first introduced in Math 21
 - D** Determine the vertex and axis of symmetry.
SK-QUAD-VERTEX · first introduced in Math 21
 - R** Recall and correctly use quadratic-function terms including parabola, vertex, axis of symmetry, intercept, zero, and solution.
SK-QUAD-VOCABULARY · first introduced in Math 21
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M31-QUAD-002

I can connect standard, factored, and vertex forms to features of a quadratic graph.

OpenStax: **Primary resources:** [5.1 Quadratic Functions](#)

Supplemental resources: [5.3 Graphs of Polynomial Functions](#)

SUPPORTING SKILLS

- D** Identify standard, factored, and vertex forms.
SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
 - A** Determine x- and y-intercepts of a quadratic function.
SK-QUAD-INTERCEPTS · first introduced in Math 21
 - A** Determine the vertex and axis of symmetry.
SK-QUAD-VERTEX · first introduced in Math 21
 - D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21
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M31-QUAD-003

I can rewrite a quadratic function in a form that reveals a requested feature.

OpenStax: **Primary resources:** [5.1 Quadratic Functions](#)

SUPPORTING SKILLS

- A** Factor a quadratic trinomial with a nonunit leading coefficient.
SK-FACTOR-TRINOMIAL-GENERAL · first introduced in Math 12
- A** Factor a monic quadratic trinomial.
SK-FACTOR-TRINOMIAL-MONIC · first introduced in Math 12
- D** Create a perfect-square trinomial by completing the square.
SK-QUAD-COMPLETE-SQUARE · first introduced in Math 21
- D** Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
SK-QUAD-FORM-CONVERT · first introduced in Math 21

M31-QUAD-004

I can solve a quadratic equation by factoring or using the quadratic formula.

OpenStax: **Primary resources:** [2.5 Quadratic Equations](#)

SUPPORTING SKILLS

- A** Apply the zero-product property.
SK-FACTOR-ZERO-PRODUCT · first introduced in Math 21
- A** Use the discriminant to predict the number and type of solutions.
SK-QUAD-DISCRIMINANT · first introduced in Math 21
- A** Substitute coefficients accurately into the quadratic formula.
SK-QUAD-FORMULA · first introduced in Math 21
- D** Select an efficient quadratic-solving method from equation structure.
SK-QUAD-METHOD-SELECT · first introduced in Math 21

M31-QUAD-005

I can graph a quadratic function using its algebraic structure and key features.

OpenStax: **Primary resources:** [5.1 Quadratic Functions](#)

SUPPORTING SKILLS

- A** Identify standard, factored, and vertex forms.
SK-QUAD-FORM-IDENTIFY · first introduced in Math 21
- D** Determine x- and y-intercepts of a quadratic function.
SK-QUAD-INTERCEPTS · first introduced in Math 21
- D** Determine the vertex and axis of symmetry.
SK-QUAD-VERTEX · first introduced in Math 21
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-QUAD-006

I can construct a quadratic function from specified zeros, a vertex, or graphical information.

OpenStax: **Primary resources:** [5.1 Quadratic Functions](#)

Supplemental resources: [5.5 Zeros of Polynomial Functions](#)

SUPPORTING SKILLS

- A** Translate quantities and relationships into an equation.
SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- I** Build a polynomial from zeros, multiplicities, and a point.
SK-POLY-CONSTRUCT
- A** Rewrite a quadratic among standard, factored, and vertex forms using an appropriate algebraic method.
SK-QUAD-FORM-CONVERT · first introduced in Math 21

M31-QUAD-007

I can create and interpret a quadratic trajectory model.

OpenStax: **Primary resources:** [5.1 Quadratic Functions](#)

SUPPORTING SKILLS

- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- D** Interpret vertex and zeros in an application.
SK-QUAD-MODEL-FEATURES · first introduced in Math 21
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

Exponential Functions and Models

REQUIRED

OpenStax coverage: **Primary resources:** [6.1 Exponential Functions](#); [6.2 Graphs of Exponential Functions](#)

M31-EXP-001

I can distinguish linear change from exponential change using differences, ratios, or percent change.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- D** Use constant differences or ratios to distinguish linear and exponential patterns.
SK-EXP-DIFFERENCE-RATIO · first introduced in Math 31
- D** Compare functions shown in different representations.
SK-FUNC-COMPARE-REP · first introduced in Math 21
- A** Calculate and interpret percent increase or decrease.
SK-NUM-PERCENT-CHANGE · inherited from middle school

M31-EXP-002

I can identify initial value and growth or decay factor in an exponential model.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- I** Identify initial amount, rate, and time interval in context.
SK-EXP-MODEL-CONTEXT
- I** Convert a percent increase or decrease to a growth or decay factor.
SK-EXP-PERCENT-FACTOR
- I** Recall and correctly use exponential-function terms including initial value, growth or decay factor, growth or decay rate, and asymptote.
SK-EXP-VOCABULARY
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21

M31-EXP-003

I can write an exponential function from a table, graph, or context.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- D** Identify initial amount, rate, and time interval in context.
SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- I** Determine an exponential model from tabular values.
SK-EXP-MODEL-TABLE
- I** Solve for an exponential model using two points.
SK-EXP-MODEL-TWO-POINTS
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-EXP-004

I can evaluate and interpret an exponential model.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- D** Identify initial amount, rate, and time interval in context.
SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- A** Identify input and output values in a formula, graph, table, or context.
SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M31-EXP-005

I can calculate growth under periodic compounding.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- I** Substitute correctly into a periodic-compounding model.
SK-EXP-COMPOUND-PERIODIC
- A** Convert a percent increase or decrease to a growth or decay factor.
SK-EXP-PERCENT-FACTOR · first introduced in Math 31
- A** Calculate a percentage of a quantity or determine what percent one quantity is of another.
SK-NUM-PERCENT-CALCULATE · inherited from middle school

M31-EXP-006

I can calculate growth under continuous compounding.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- I** Substitute correctly into a continuous-compounding model.
SK-EXP-COMPOUND-CONTINUOUS
- A** Identify initial amount, rate, and time interval in context.
SK-EXP-MODEL-CONTEXT · first introduced in Math 31

M31-EXP-007

I can determine an exponential formula from two points or other sufficient data.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- A** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- D** Solve for an exponential model using two points.
SK-EXP-MODEL-TWO-POINTS · first introduced in Math 31
- A** Interpret numerator and denominator roles in a rational exponent.
SK-EXP-RATIONAL · first introduced in Math 21

M31-EXP-008

I can determine domain, range, intercepts, and asymptotes of an exponential function.

OpenStax: **Primary resources:** [6.2 Graphs of Exponential Functions](#)

SUPPORTING SKILLS

- I** Identify horizontal asymptotic behavior of an exponential graph.
SK-EXP-ASYMPTOTE
- D** Determine domain restrictions from denominators, even roots, and logarithms.
SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-EXP-009

I can compare the growth rates of two exponential functions.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- I** Compare exponential growth rates expressed in different forms.
SK-EXP-COMPARE-GROWTH
- D** Compare functions shown in different representations.
SK-FUNC-COMPARE-REP · first introduced in Math 21

M31-EXP-010

I can determine the percent growth or decay represented by a continuous exponential model.

OpenStax: **Primary resources:** [6.1 Exponential Functions](#)

SUPPORTING SKILLS

- I** Convert between equivalent discrete- and continuous-growth forms.
SK-EXP-DISCRETE-CONTINUOUS
- D** Convert a percent increase or decrease to a growth or decay factor.
SK-EXP-PERCENT-FACTOR · first introduced in Math 31
- A** Calculate and interpret percent increase or decrease.
SK-NUM-PERCENT-CHANGE · inherited from middle school

Logarithmic Functions and Models

REQUIRED

OpenStax coverage: **Primary resources:** [6.3 Logarithmic Functions](#); [6.5 Logarithmic Properties](#); [6.6 Exponential and Logarithmic Equations](#); [6.7 Exponential and Logarithmic Models](#); [6.4 Graphs of Logarithmic Functions](#)
Supplemental resources: [6.2 Graphs of Exponential Functions](#)

M31-LOG-001

I can convert between logarithmic and exponential forms.

OpenStax: **Primary resources:** [6.3 Logarithmic Functions](#)

SUPPORTING SKILLS

- I** Convert between logarithmic and exponential forms.
SK-LOG-CONVERT
- I** Interpret a logarithm as an exponent.
SK-LOG-DEFINITION
- I** Read logarithmic notation and identify the base, argument, and corresponding exponential statement.
SK-LOG-NOTATION-READ

M31-LOG-002

I can evaluate a logarithm using its exponential meaning.

OpenStax: **Primary resources:** [6.3 Logarithmic Functions](#)

SUPPORTING SKILLS

- I** Rewrite exponential expressions using a common base when possible.
SK-EXP-REWRITE-BASE
- D** Interpret a logarithm as an exponent.
SK-LOG-DEFINITION · first introduced in Math 31

M31-LOG-003

I can expand or condense an expression using product, quotient, and power properties.

OpenStax: **Primary resources:** [6.5 Logarithmic Properties](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I** Apply the logarithm power property in either direction.
SK-LOG-POWER
- I** Apply the logarithm product property in either direction.
SK-LOG-PRODUCT
- I** Apply the logarithm quotient property in either direction.
SK-LOG-QUOTIENT

M31-LOG-004

I can solve an exponential equation using logarithms.

OpenStax: **Primary resources:** [6.6 Exponential and Logarithmic Equations](#)

SUPPORTING SKILLS

- A** Isolate a specified variable in an equation or formula.
SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- A** Apply the logarithm power property in either direction.
SK-LOG-POWER · first introduced in Math 31
- I** Isolate an exponential expression and apply a logarithm.
SK-LOG-SOLVE-EXP

M31-LOG-005

I can solve a logarithmic equation using exponential form and check domain restrictions.

OpenStax: **Primary resources:** [6.6 Exponential and Logarithmic Equations](#)

SUPPORTING SKILLS

- A** Convert between logarithmic and exponential forms.
SK-LOG-CONVERT · first introduced in Math 31
- I** Reject solutions that make a logarithm argument nonpositive.
SK-LOG-DOMAIN-CHECK
- I** Combine or isolate logarithms before converting to exponential form.
SK-LOG-SOLVE-LOG

M31-LOG-006

I can calculate and interpret doubling time or half-life.

OpenStax: **Primary resources:** [6.7 Exponential and Logarithmic Models](#)

SUPPORTING SKILLS

- I** Translate a doubling-time or half-life question into an exponential equation.
SK-LOG-DOUBLING-HALF
- A** Isolate an exponential expression and apply a logarithm.
SK-LOG-SOLVE-EXP · first introduced in Math 31
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

M31-LOG-007

I can determine domain, range, intercepts, and asymptotes of a logarithmic function.

OpenStax: **Primary resources:** [6.4 Graphs of Logarithmic Functions](#)

SUPPORTING SKILLS

- D** Determine domain restrictions from denominators, even roots, and logarithms.
SK-FUNC-DOMAIN-FORMULA · first introduced in Math 31
- I** Identify vertical asymptotic behavior of a logarithmic graph.
SK-LOG-ASYMPTOTE
- A** Reject solutions that make a logarithm argument nonpositive.
SK-LOG-DOMAIN-CHECK · first introduced in Math 31
- D** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-LOG-008

I can explain how exponential and logarithmic functions undo one another.

OpenStax: **Primary resources:** [6.3 Logarithmic Functions](#)

SUPPORTING SKILLS

- A** Exchange input and output values when interpreting an inverse.
SK-FUNC-INVERSE-SWAP · first introduced in Math 31
- A** Verify inverse functions using composition.
SK-FUNC-INVERSE-VERIFY · first introduced in Math 31
- D** Interpret a logarithm as an exponent.
SK-LOG-DEFINITION · first introduced in Math 31

M31-LOG-009

I can solve a multi-step exponential or logarithmic equation.

OpenStax: **Primary resources:** [6.6 Exponential and Logarithmic Equations](#)

SUPPORTING SKILLS

- A** Reject solutions that make a logarithm argument nonpositive.
SK-LOG-DOMAIN-CHECK · first introduced in Math 31
- A** Apply the logarithm power property in either direction.
SK-LOG-POWER · first introduced in Math 31
- A** Apply the logarithm product property in either direction.
SK-LOG-PRODUCT · first introduced in Math 31
- A** Apply the logarithm quotient property in either direction.
SK-LOG-QUOTIENT · first introduced in Math 31
- D** Isolate an exponential expression and apply a logarithm.
SK-LOG-SOLVE-EXP · first introduced in Math 31
- D** Combine or isolate logarithms before converting to exponential form.
SK-LOG-SOLVE-LOG · first introduced in Math 31

M31-LOG-010

I can determine an intersection of exponential functions graphically or algebraically.

OpenStax: **Primary resources:** [6.6 Exponential and Logarithmic Equations](#)

Supplemental resources: [6.2 Graphs of Exponential Functions](#)

SUPPORTING SKILLS

- A** Compare exponential growth rates expressed in different forms.
SK-EXP-COMPARE-GROWTH · first introduced in Math 31
- D** Isolate an exponential expression and apply a logarithm.
SK-LOG-SOLVE-EXP · first introduced in Math 31
- A** Check an ordered pair in both equations of a system.
SK-SYS-CHECK · first introduced in Math 12
- A** Locate and interpret the intersection of two lines.
SK-SYS-GRAPH-INTERSECTION · first introduced in Math 12

M31-LOG-011

I can rewrite a discrete exponential model as an equivalent continuous exponential model.

OpenStax: **Primary resources:** [6.7 Exponential and Logarithmic Models](#)

SUPPORTING SKILLS

- D** Convert between equivalent discrete- and continuous-growth forms.
SK-EXP-DISCRETE-CONTINUOUS · first introduced in Math 31
- A** Convert a percent increase or decrease to a growth or decay factor.
SK-EXP-PERCENT-FACTOR · first introduced in Math 31
- A** Isolate an exponential expression and apply a logarithm.
SK-LOG-SOLVE-EXP · first introduced in Math 31

I can solve and interpret an exponential or logarithmic modeling problem.

OpenStax: **Primary resources:** [6.7 Exponential and Logarithmic Models](#)

SUPPORTING SKILLS

- D** Identify initial amount, rate, and time interval in context.
SK-EXP-MODEL-CONTEXT · first introduced in Math 31
- A** Isolate an exponential expression and apply a logarithm.
SK-LOG-SOLVE-EXP · first introduced in Math 31
- A** Combine or isolate logarithms before converting to exponential form.
SK-LOG-SOLVE-LOG · first introduced in Math 31
- D** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A** Restrict a model to meaningful contextual inputs.
SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A** Attach and interpret units for rates, inputs, and outputs.
SK-REP-UNITS · first introduced in Math 12

Extension Content

EXTENSION

OpenStax coverage: **Primary resources:** [3.6 Absolute Value Functions](#); [3.5 Transformation of Functions](#); [6.7 Exponential and Logarithmic Models](#)

M31-EXT-001

I can analyze an absolute-value function using transformations.

OpenStax: **Primary resources:** [3.6 Absolute Value Functions](#)

SUPPORTING SKILLS

- A** Interpret transformations applied to function inputs.
SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31
- A** Distinguish stretches from compressions and horizontal effects from vertical effects.
SK-FUNC-TRANSFORM-SCALE · first introduced in Math 31
- A** Interpret transformations applied to function outputs.
SK-FUNC-TRANSFORM-VERTICAL · first introduced in Math 31
- A** Connect parameters in a formula to features of its graph.
SK-REP-FORMULA-GRAPH · first introduced in Math 21

M31-EXT-002

I can determine even or odd symmetry from a graph or formula.

OpenStax: **Primary resources:** [3.5 Transformation of Functions](#)

SUPPORTING SKILLS

- A** Simplify an algebraic expression using operation properties.
SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I** Test a formula or graph for even or odd symmetry.
SK-FUNC-SYMMETRY

M31-EXT-003

I can interpret a contextual logarithmic scale.

OpenStax: **Primary resources:** [6.7 Exponential and Logarithmic Models](#)

SUPPORTING SKILLS

- A** Interpret a logarithm as an exponent.
SK-LOG-DEFINITION · first introduced in Math 31
- I** Interpret equal intervals on a logarithmic scale as equal multiplicative changes.
SK-LOG-SCALE-INTERPRET
- A** Explain a model parameter in the language of its context.
SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21